(M___.0) ANTI-PLASMA MINE

Once the Plasma Sabot (FP11.0) became widely-deployed, the Federation Star Fleet, recognizing the growing threat, crash-restarted a previously-attempted program. The hope was that decades of scientific advances since the original effort had been shelved would enable a workable solution to deal with the new technology. The Anti-Plasma Mine (APM) was the end result; it was specifically designed to defend against plasma torpedoes. By mid-Y182, Star Fleet hastily deployed advanced prototypes to selected bases on the Romulan front. Military secrets being the most fleeting of all, other Galactic empires quickly stole the designs and developed copycat mines by late Y183.

While APMs leveraged many of the same components as typical explosive mines, the plasma detection technologies used for the triggering mechanisms were quite new. The complexity of the triggering systems made reliably scaling production very difficult; moreover, even commanders defending against plasma-armed enemies were reluctant to place a great deal of faith in the new technology, and the limits on their effectiveness beyond plasmas meant no one else particularly wanted them. As a consequence, APMs were never widely used.

The Andromedans were first observed to use a functionally-similar device in early Y193, but only did so rarely, preferring PA Mines (M10.0) instead. Galactic scientists hypothesized that while the Andromedans clearly understood how APMs worked, they were unable, or unwilling, to produce significant quantities of them. There is no record of empires outside the Alpha Octant using APM-like mines.

(M___.1) DEPLOYMENT

(M___.11) PERMITTED UNITS: Any unit normally able to carry mines (or T-bombs) may carry an Anti-Plasma Mine (APM).

(M___.12) MINE REPLACED: An APM replaces a mine (or T-bomb) the unit would normally carry.

(M___.121) When an APM replaces a T-bomb, the "dummy" (M2.9) T-bomb is retained and may be used normally.

(M___.122) There is no "dummy" version of an APM.

(M___.13) SIZES, STRENGTH: The small APM has the same mass and form-factor as a T-bomb (M3.0), and a Degradation Strength (M___.33) of seven; the large APM has the same mass and form-factor as a Nuclear Space Mine (M2.7), and a Degradation Strength of 21.

(M___.14) COST: An APM costs one BPV more than a conventional explosive mine of the same size (M6.3).

(M___.15) AVAILABILITY: The first year this system is available to an empire, a non-minelaying ship that has a normal explosive mine carriage capability of five or more may replace no more than one conventional explosive mine with an APM; bases may replace up to two explosive mines, either one of each size or two of the same size. See (M___.5) for Mine Fields (M6.0).

(M___.151) Mine-layers (M9.0) may substitute APMs for up to 20% (round down) of their total explosive mine-carrying capacity during the first year APMs are available.

(M___.152) Mine-capable shuttles (such as an MLS) may not have an APM in their own equipment store, but may carry APMs drawn from their home ship's mine capacity.

(M___.153) One year after this system first becomes available, a non-minelaying ship may replace no more than 25% (round down, but not less than one) of its total conventional mines with APMs; bases and mine-layers may replace up to 50% of total conventional mines. However, this does not apply to Andromedan units, which are always treated as if it were the first year of availability.

(M___.154) Even after the first year of availability, a non-base unit may not replace more than 50% (round down, but not less than one) of its normal mine complement with APMs. While units with cargo that purchase equipment under (G25.43) may purchase additional APMs, the total number of APMs aboard a non-base unit cannot exceed 50% (round down) of all mines carried by the unit. A base may have up to 70% (round down) of its explosive mines aboard be APMs, but the APMs over 50% must be in cargo.

(M___.16) CONDITIONS: APMs are subject to (M2.2).

(M___.17) MINESWEEPING: The damage required to destroy an APM is the same as for a conventional explosive mine of the same size (M8.41); APMs may be triggered under (M8.42) as if an explosive mine.

(M___.18) TYPE NOT KNOWN: The nature of an APM (*e.g.,* that it is not a conventional explosive mine) is not detectable except by (M7.5), or when it is triggered.

(M___.19) BOLTS, CARRONADES NOT AFFECTED: APMs do not interact with Plasma Bolts (FP8.0) or Plasma Carronades (FP14.0). Bolts and carronade shots are not detected by, and cannot trigger, APMs.

(M___.2) OPERATION

(M___.21) PLACEMENT: APMs of any size may be placed using (M2.1); a small APM may be placed using (M3.2).

(M___.211) An APM is subject to (M2.6) if explosive mines are also subject.

(M___.212) If used in a Mine Field (M6.0), see (M___.5).

(M___.22) ARMING: APMs arm under the same conditions (M2.3) (M3.223) as conventional explosive mines.

(M___.23) DETECTION CRITERIA: By default, an APM accepts any plasma torpedo, including ECM Plasmas (FP12.0), as a valid target (this creates a partial exception to (FP1.84)), regardless of the strength of the warhead. Players have some control over what targets the mine will accept.

(M___.231) An APM may be set to detect a plasma torpedo only if the warhead strength (FP1.5) meets or exceeds a specific minimum value; this is similar to (M2.14). Warhead strength after movement should be used; if the minimum value is not met/exceeded, the plasma torpedo is ignored and also would not count for the purposes of (M___.232).

> **(M___.2311)** A PPT is detected as a valid target if the simulated warhead strength (FP6.3) meets or exceeds the detection criteria.

(M___.2312) If multiple plasma torpedoes move within the detection zone (M2.41), do not add their warhead strengths to determine if the minimum value has been met; each plasma torpedo is evaluated individually.

(M___.2313) An ECM Plasma (FP12.0) is ignored (as if the warhead strength detection criteria was not met) if an APM has been configured to require a minimum warhead strength, but can still trigger an APM that does not use (M___.231).

(M___.232) A mine may be set to ignore up to six plasma torpedoes, either with or without regard to warhead strength, similar to (M2.15).

(M___.2321) An ECM Plasma (FP12.0) is detected and would count as a valid target if (M___.231) is not in use.

(M___.2322) A PPT is detected as a valid target; if (M___.231) is used, then the simulated warhead strength (FP6.3) must also meet/exceed the detection criteria (M___.231).

(M___.24) TRIGGERING: An armed APM is triggered using the same criteria as a conventional explosive mine, except that only a plasma torpedo may trigger an APM; no other unit may do so.

(M___.241) A plasma torpedo still must meet the (M___.23) detection criteria.

(M___.242) If multiple plasma torpedoes enter the detection zone (M2.41) of an APM, use (M2.47) to resolve, treating each individual plasma torpedo as a discrete unit.

(M___.3) EFFECT ON PLASMA TORPEDOES

(M___.31) MECHANISM OF ACTION: When triggered, an APM briefly generates a hyper-intense magnetic pulse, specifically tuned such that it degrades the magnetic field used to contain the warhead of a plasma torpedo. The effects of an APM form an exception under (FP1.62).

(M___.32) ZONE OF EFFECT: An APM has a zone of effect (similar to the "explosion zone") equal to that of a conventional explosive mine (M2.50).

(M___.321) This zone is the same regardless of APM size.

(M___.322) The effects of the APM are applied to all in-flight plasma torpedoes in the zone of effect at the moment the APM is triggered, regardless of launcher or intended target (*e.g.,* the APM may affect plasma torpedoes launched by the same side that laid the mine).

(M___.33) DEGRADATION: Each plasma torpedo in the zone of effect is affected as if it had moved a number of hexes equal to the Degradation Strength (M___.13) of the mine; exception (M___.333). The effect is permanent (*e.g.,* the endurance of the torpedo is reduced), although there may be no immediate reduction in the warhead strength.

(M___.331) This effect is applied instantly to the plasma torpedo, and if the warhead strength is reduced to zero or below, the torpedo is immediately removed (FP1.51).

(M___.332) PPTs are affected as if they were normal plasma torpedoes (*e.g.,* the deception of a PPT is not revealed by an APM).

(M___.333) If an affected plasma torpedo is an ECM Plasma (FP12.0), then multiply the Degradation Strength (M___.13) of the APM by three and reduce the endurance of the ECP (FP12.23) by that many impulses.

(M___.34) AFTER IMPACT: In the Sequence of Play, the effects of mines are resolved after the effects of seeking weapon impacts, so a plasma torpedo that impacts its target does so before an APM can trigger; however, any such plasma torpedo cannot trigger an APM and would not count as a valid target for the purposes of (M___.23).

(M___.35) BEFORE LAUNCH: In the Sequence of Play, the effects of mines are resolved prior to seeking weapon launches; an APM pulse has no effect on unlaunched plasma torpedoes held in launch tubes (or canisters) aboard ships or shuttles or captor mines in the zone of effect; the plasma torpedo systems aboard those units are capable of keeping the warheads stable.

(M___.4) SECONDARY EFFECTS OF ANTI-PLASMA MINES

(M___.41) EW-LIKE EFFECTS: When an APM is triggered, the electro-magnetic pulse generated is so intense that units within the zone of effect (including in-flight plasma torpedoes) receive EW points equal to the square root (round down) of the Degradation Strength (M___.13) of the APM, as follows:

(M___.411) ECM, from a "natural" (D6.3143) source; and

(M___.412) Offensive ECM (D6.3145).

(M___.413) These effects are cumulative (M___.43).

(M___.42) LIMITED EFFECTS DURATION: The (M___.41) EW-like effects last a very short time that varies by the affected unit:

(M___.421) All units subject to (M___.411) are affected until the start of the POST COMBAT SEGMENT (6E) of the impulse in which the APM was triggered.

(M___.422) Plasma torpedoes subject to (M___.412) are affected until the MINE DAMAGE RESOLUTION STEP of DAMAGE DURING MOVEMENT (6A3) of the impulse <u>after</u> the APM was triggered; all other units are affected until the start of the POST COMBAT SEGMENT (6E) of the impulse in which the APM was triggered.

(M___.43) MULTIPLE MINES: If multiple APMs are triggered and cause EW-like effects (M___.41), the effects are cumulative for units that are within the intersection of zones of effect (that is, when calculating, add the effects of each APM; do not add the Degradation Strengths of the APMs and then take the square root). The duration of those effects (M___.42) is not increased, regardless of how many APMs are triggered in one impulse, nor will any unit lose lock-on due solely to those effects.

(M___.44) CLOAKING DEVICE: The (M___.41) effects interact with a Cloaking Device (G13.0) as follows:

(M___.441) APMs do not cause the "flash-bulb effect" of explosive mines and do not void a cloak under (G13.52).

(M___.442) If a unit is attempting to retain lock-on against a cloaked unit that is subject to (M___.41), then the ECM provided to the cloaked unit under (M___.411) may require a new die roll (G13.332).

(M___.443) If a unit attempting to retain lock-on against a cloaked unit is itself subject to (M___.41), then the Offensive EW (M___.412) applied to it may require a new die roll (G13.332).

(M___.444) At most, the (M___.41) effects can require a unit attempting to acquire/retain lock-on against a cloaked unit to make a single (G13.52) new die roll, regardless of how many APMs are triggered in one impulse. While the effects of multiple APMs are cumulative (M___.43), each APM triggered does not require a separate roll (however, the cumulative effects may make it practically impossible for the unit to acquire/retain lock-on).

(M___.45) WEBS: The effects of an APM are not diminished by a web, so (G10.76) does not apply; webs are themselves not affected by the magnetic pulse from an APM.

(M___.46) ESGS: ESG fields are not damaged by APM bursts, but APMs are themselves damaged/destroyed by an ESG (G23.612), and trigger if an explosive mine would trigger under (G23.6).

(M___.47) SPECIAL SENSORS: The magnetic energy burst from one or more APMs will not blind a powered Special Sensor channel on a unit within the zone of effect; however, in and of themselves, Special Sensors do not overcome the EW-like effects (M___.41) to which the Special Sensor-equipped unit is otherwise subject (*e.g.,* just because a unit has Special Sensors doesn't allow it to ignore (M___.412)).

(M___.48) TACTICAL INTELLIGENCE: When an APM is triggered, it is detectable at Level A (D17.3).

(M___.49) NO DAMAGE OR OTHER EFFECTS: APMs do not damage any unit other than plasma torpedoes, unless specifically stated to do so in other rules. Andromedan PA panels ignore the generated magnetic pulse (no power is absorbed). Absent a specific rule to the contrary, APMs do not trigger or disable other mines (including plasma Captor mines), break seeking weapon locks, distract seeking weapons, or disable any other unit or weapon.

(M___.5) MINE FIELDS

(M___.51) EXPLOSIVE MINES REPLACED: Mine Fields (M6.0) may substitute one APM for an explosive mine, of the same size, for the cost differential stated (M___.14); and, may do so for each size of explosive mine in the field.

(M___.511) Typically, most Mine Fields would not replace more than one explosive mine, of a given size, with an APM. However, bases defending areas with plasma-armed enemies may replace up to three explosive mines, of a given size, with APMs of the same size as the replaced mine.

(M___.512) These limits are *per* Mine Field, not *per* base or location.

(M___.52) DEPLOYMENT AND MAINTENANCE: APMs in Mine Fields are deployed, maintained and replaced in the same manner as any other mines in the Mine Field.

(M___.53) CONTROL: APMs in a Mine Field may be command-controlled (M5.2) by a base.